

Paper:

Enhancing Operation Wisdom Logging Solution Utilizing Different Local LLMs for Security Anomaly Detection

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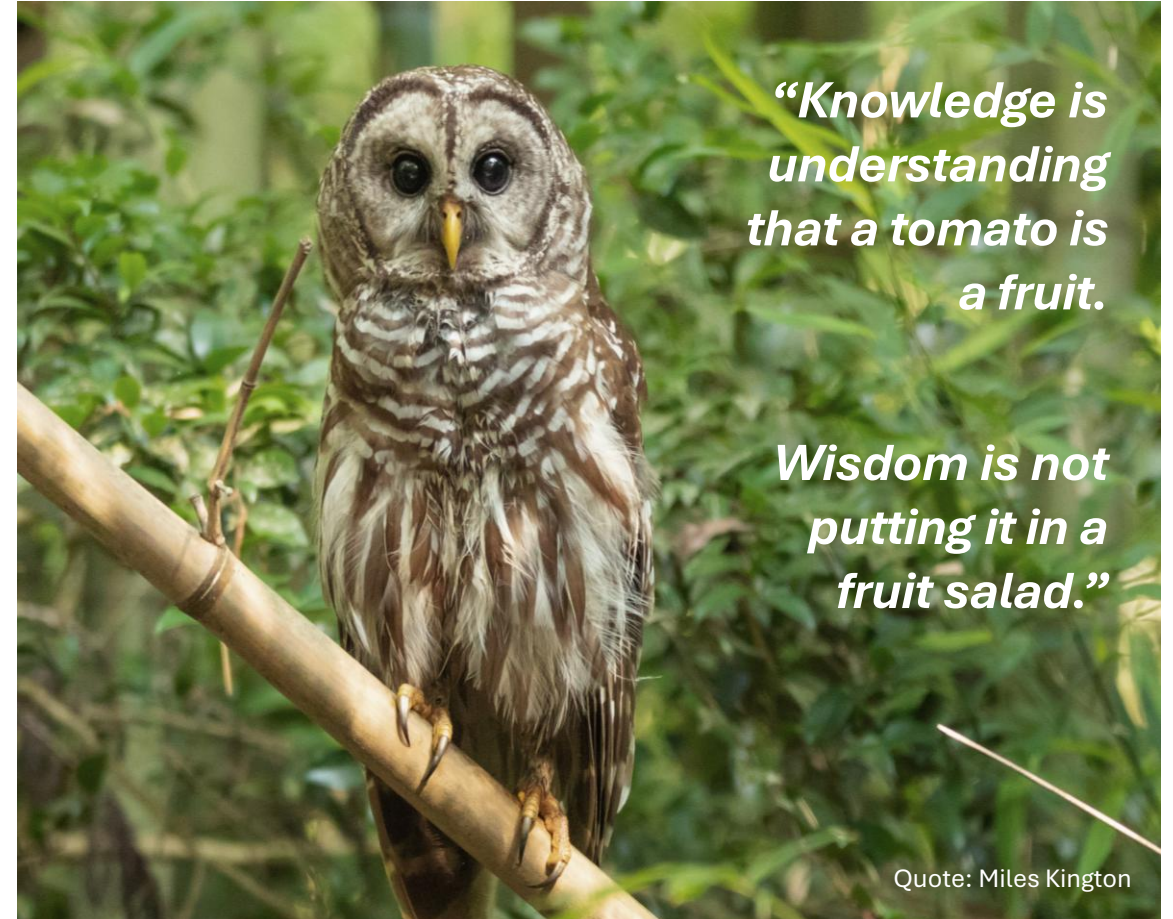
This is a story...

...how to
monetize
data with
OWL?

Operations

Wisdom

Logging



Fragmented, Reactive, and Unaudited Operations of End Nodes

End Node Definition

Any IT, OT, or Telco hardware or software that requires remote access via multiple protocols.

Ericsson Nikola Tesla



Complicated
Access to the
end Node



Passive Data



Lack of
Integration



Reactive
manual
support

Integration
Teams

Competence Domain
Teams

End Node
Owners



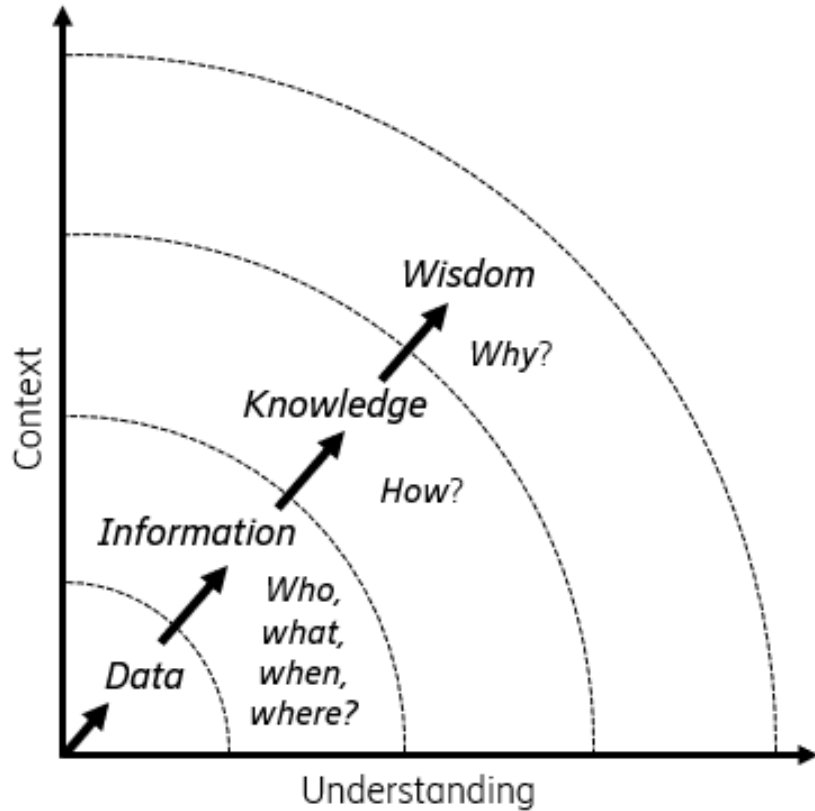
Data &
AI/ML Teams

Security Teams &
Auditors

Operational Assurance
L1-L3

ENT

Data Monetization through Unified, Secure, and Proactive Operations



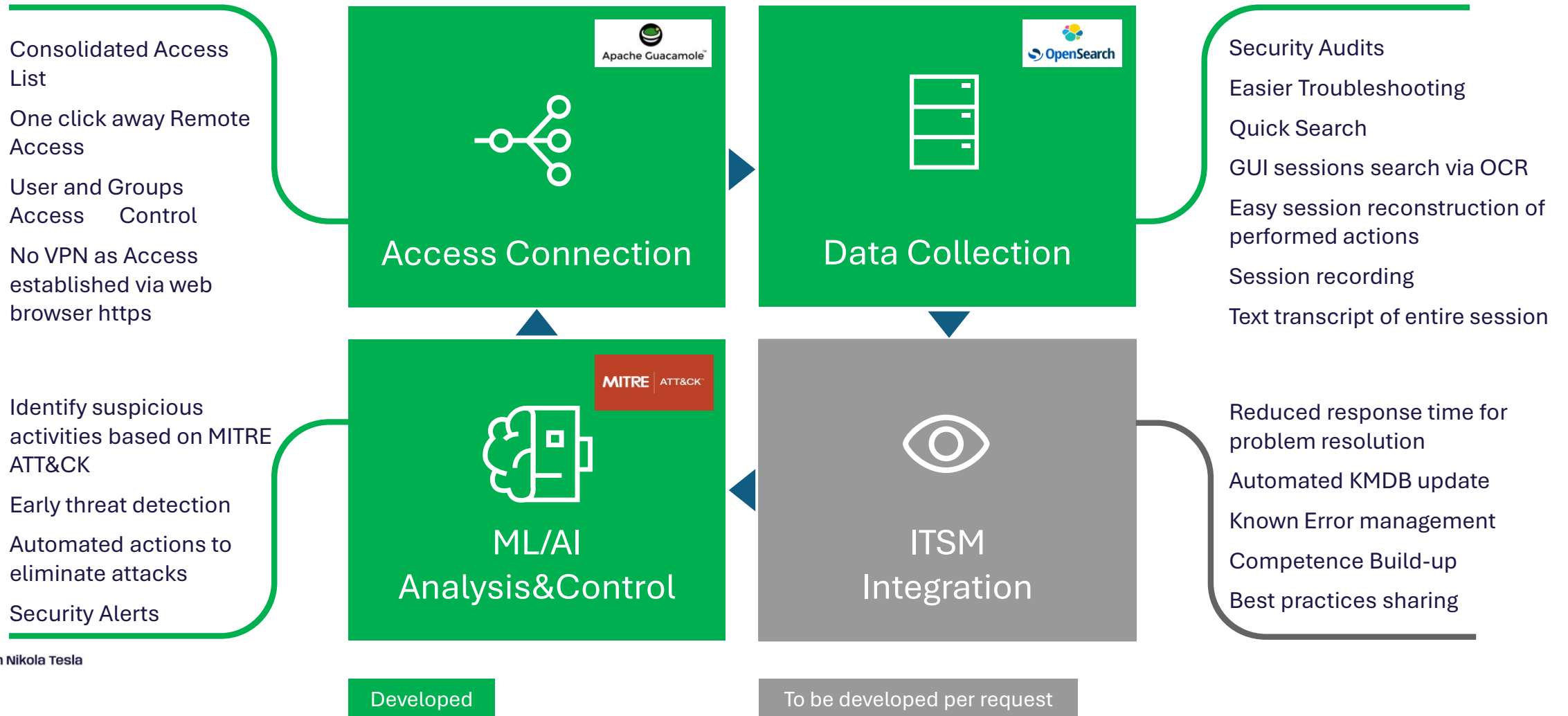
Operations Wisdom Logging (OWL) Baseline

DIKW Pyramid represents relationships between data, information, knowledge and wisdom.

OWL toolset supports data **transformation** into **actionable knowledge**, enabling informed decision-making and **value creation** through data monetization.

OWL Solution

Basic Components & Benefits



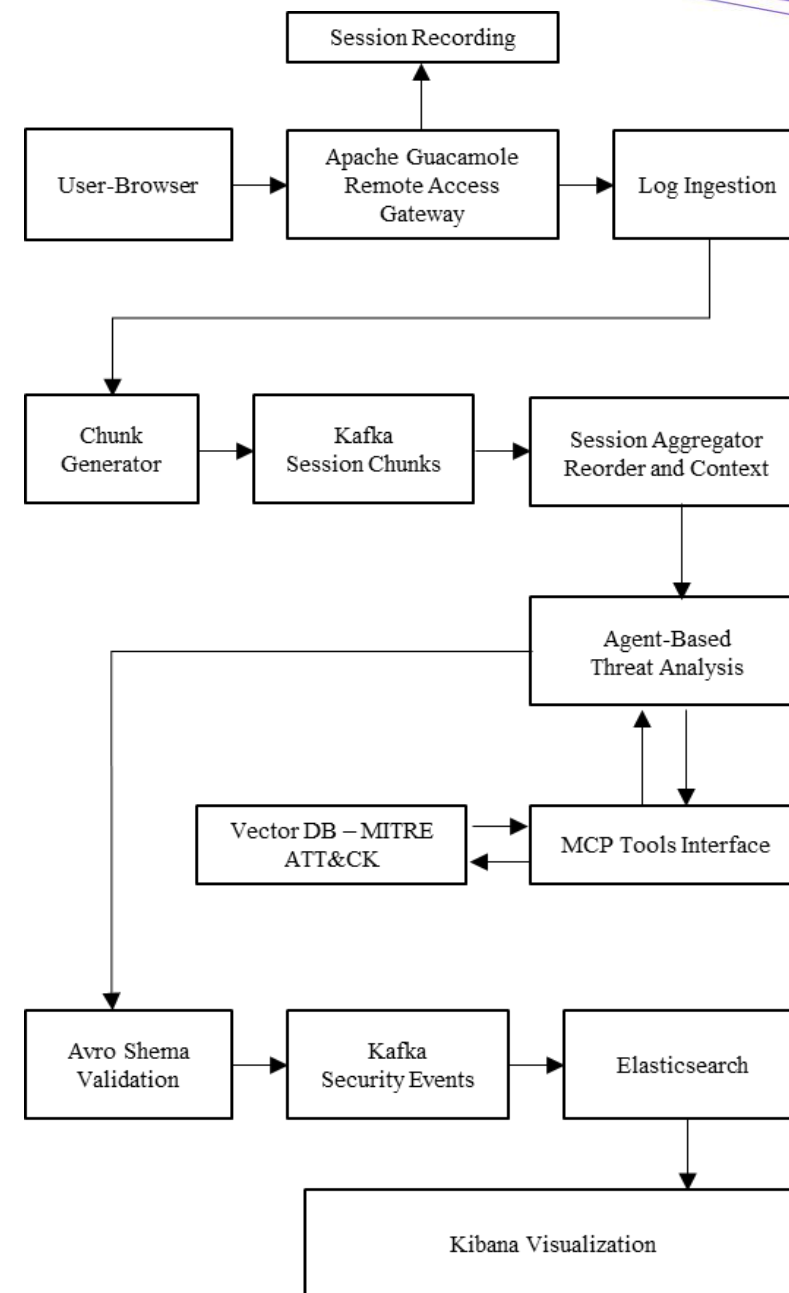
Security Analysis Pipeline

Hardware:

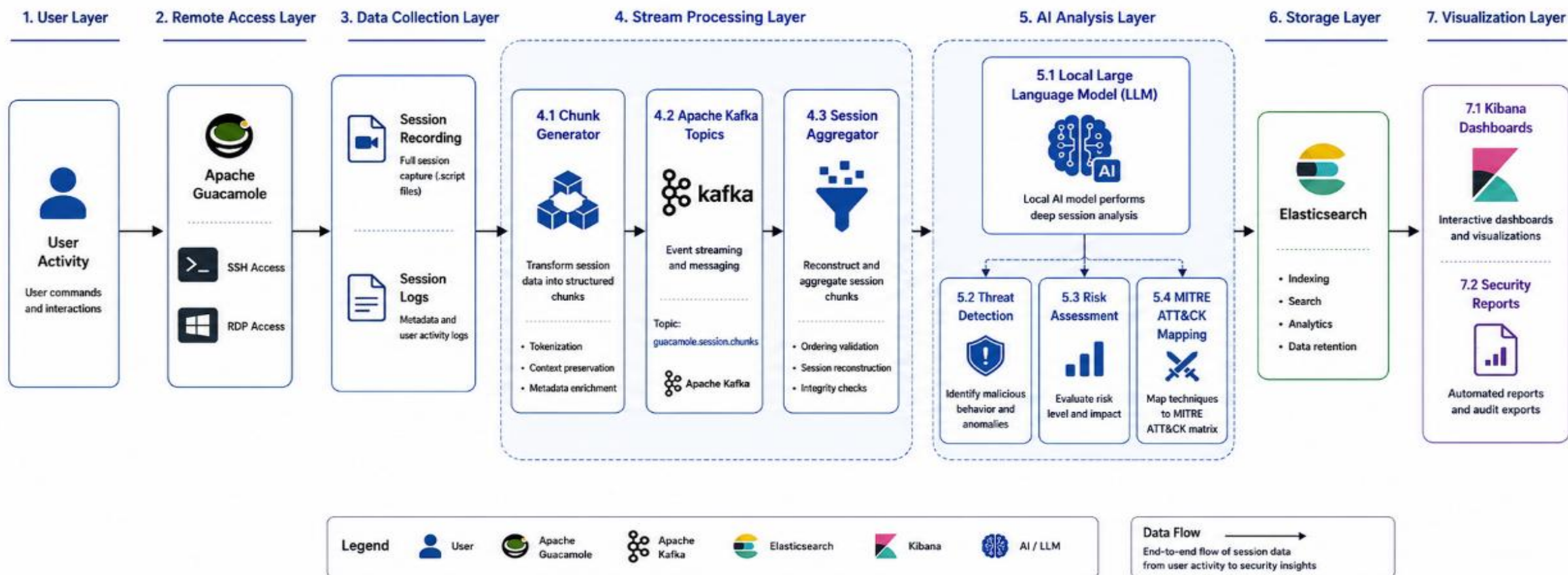
- HPE DL360 Gen10
- CPU: 2× Intel Xeon Gold 5317 @ 3.00 GHz (24 cores, 48 threads)
- RAM: 128 GB DDR4
- GPU: NVIDIA L4 (24 GB GDDR6, Ada Lovelace architecture)

LLM Model:

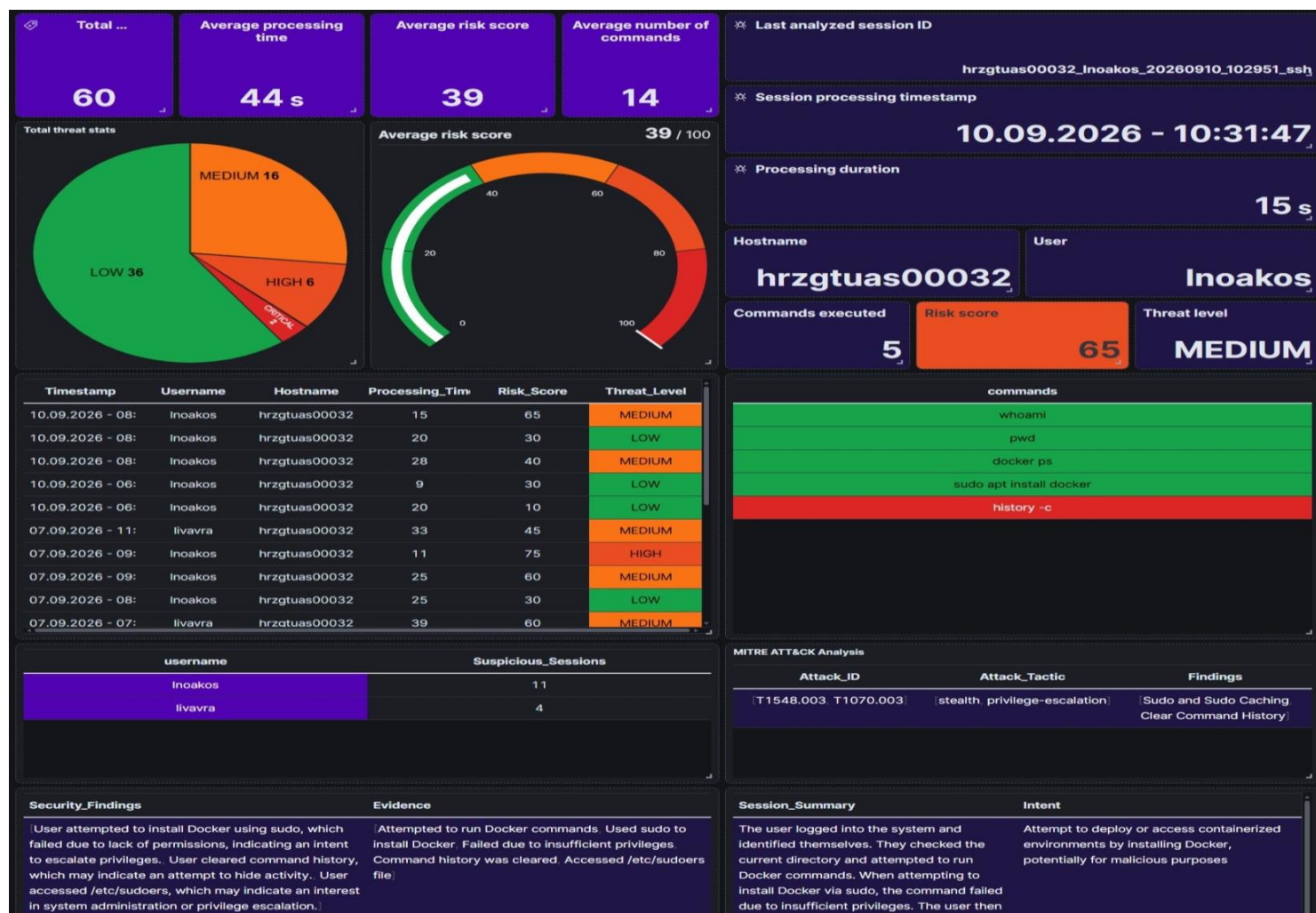
- Qwen2.5-14B-Instruct, a 14.8 billion parameter decoder-only transformer model
- The model uses Grouped Query Attention (GQA)
- Model inference is served by Ollama
- The application layer communicates through LiteLLM 1.52.9
- MITRE ATT&CK knowledge base are mediated via the Model Context Protocol (MCP)



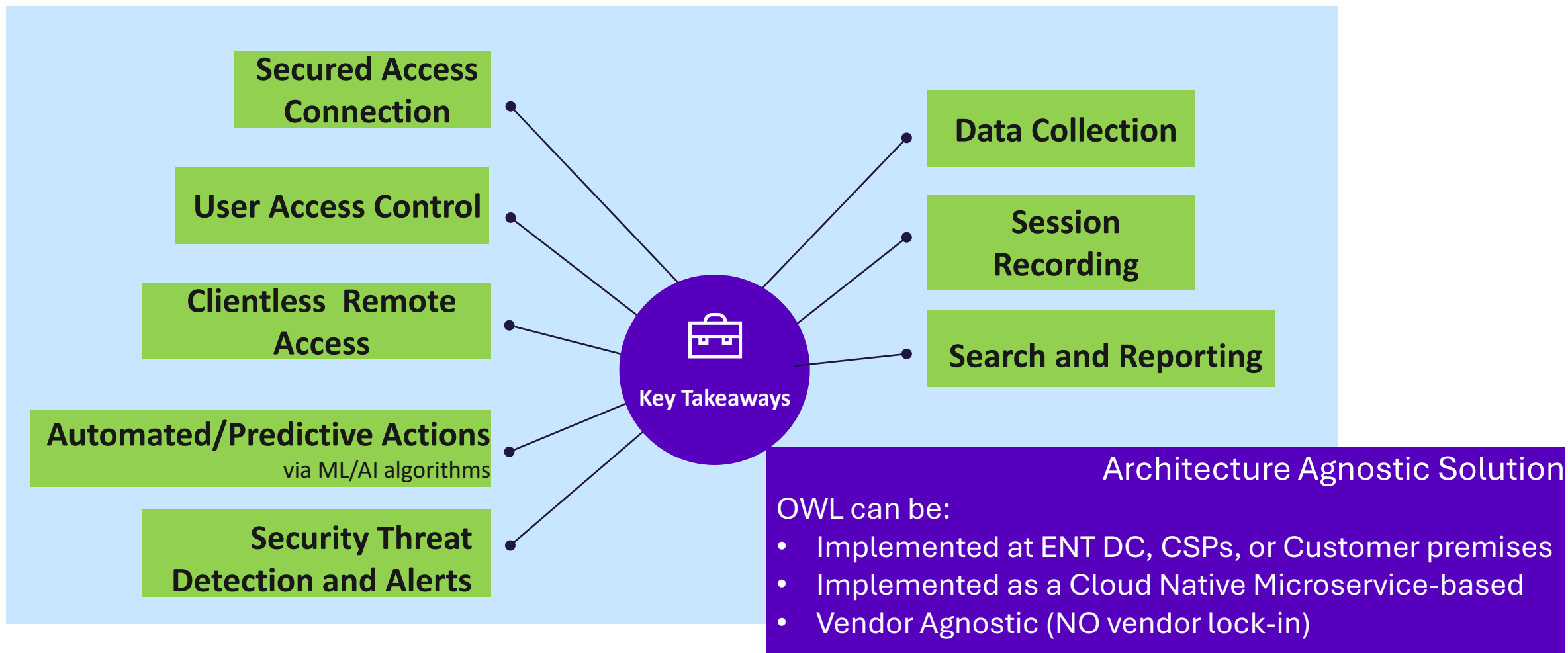
Security Pipeline Architecture



Security Anomaly Detection Automation & Reporting



OWL in ONE



A series of thin, light blue lines radiating from the top right corner of the slide, creating a sense of motion or connectivity.

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